
DSC5001: Statistical Machine Learning I

Week 4: Regression I: Linear Regression

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Learning objectives

1. Build and interpret linear regression models.
2. Derive ordinary least squares and connect it to Gaussian maximum likelihood.
3. Quantify uncertainty in coefficients and predictions.
4. Diagnose model assumptions and evaluate generalization.

The Regression Problem

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Regression: What Is the Task?

Suppose a shop records advertising spending and weekly sales.

Advertising x	1	2	3	4
Sales y	8	10	13	16

The observations form the training data:

$$\mathcal{D} = \{(x_1, y_1), \dots, (x_n, y_n)\}.$$

The task is to predict the response for a new input:

$$x_0 = 5 \quad \longrightarrow \quad \hat{y}_0 ?$$

Regression predicts a numerical response from observed features.

What Does Regression Learn?

The same input does not always produce exactly the same response.
For example, at one advertising level we might observe:

$$Y = 10, 12, 11, 13, \dots$$

Regression models the conditional mean:

$$m(\mathbf{x}) = E[Y \mid \mathbf{X} = \mathbf{x}]$$

and writes:

$$Y = m(\mathbf{X}) + \varepsilon, \quad E[\varepsilon \mid \mathbf{X}] = 0.$$

For example, if $m(5) = 18$, an individual future response might still be 16, 18, or 21.

Simple Linear Regression

With one feature x , assume the conditional mean is a straight line:

$$E[Y \mid X = x] = \beta_0 + \beta_1 x$$

β_0 expected response when $x = 0$

β_1 expected response change for a one-unit increase in x

Example:

$$\hat{y} = 5 + 2.5x.$$

At $x = 4$:

$$\hat{y} = 5 + 2.5(4) = 15.$$

So increasing x by one unit increases the fitted mean response by 2.5 units.

From One Feature to Many

Most prediction problems involve several features.

For example:

house price $\leftarrow$ {area, age, distance}.

With p features:

$$E[Y \mid \mathbf{X} = \mathbf{x}] = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$$

For observation i :

$$y_i = \beta_0 + \sum_{j=1}^p \beta_j x_{ij} + \varepsilon_i.$$

The same linear-regression idea now uses several inputs instead of one.

What Does “Linear” Mean?

Linear regression means linear in the unknown coefficients.

For example:

$$E[Y \mid X = x] = \beta_0 + \beta_1 x + \beta_2 x^2$$

is still linear in:

$$\beta_0, \beta_1, \beta_2.$$

Define:

$$\phi(x) = \begin{bmatrix} 1 \\ x \\ x^2 \end{bmatrix}.$$

Then:

$$E[Y \mid X = x] = \phi(x)^\top \beta.$$

The prediction may be nonlinear in x while remaining linear in the coefficients.

Matrix Form

Include the intercept in each feature vector:

$$\mathbf{x}_i = (1, x_{i1}, \dots, x_{ip})^\top.$$

Stack all observations:

$$\mathbf{X} = \begin{bmatrix} 1 & x_{11} & \cdots & x_{1p} \\ 1 & x_{21} & \cdots & x_{2p} \\ \vdots & \vdots & & \vdots \\ 1 & x_{n1} & \cdots & x_{np} \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}.$$

The model becomes:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$$

Population Model vs. Fitted Model

The population model contains unknown coefficients:

$$m(\mathbf{x}) = \mathbf{x}^\top \boldsymbol{\beta}.$$

The training data are used to estimate them:

$$\boldsymbol{\beta} \longrightarrow \hat{\boldsymbol{\beta}}.$$

The fitted regression function is:

$$\hat{y}(\mathbf{x}) = \mathbf{x}^\top \hat{\boldsymbol{\beta}}$$

For training observation i :

$$\hat{y}_i = \mathbf{x}_i^\top \hat{\boldsymbol{\beta}}.$$

$\boldsymbol{\beta}$		unknown population parameter
$\hat{\boldsymbol{\beta}}$		
		estimate from the data

Fitted Values and Residuals

For observation i :

$$e_i = y_i - \hat{y}_i$$

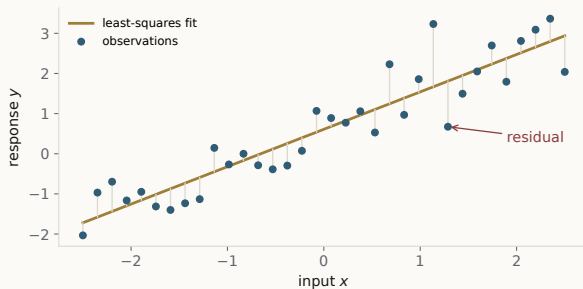
Example:

$$y_i = 18, \quad \hat{y}_i = 15$$

so:

$$e_i = 3.$$

$$\begin{array}{l|l} e_i > 0 & \text{fit is too low} \\ e_i < 0 & \text{fit is too high} \end{array}$$



Why Not Add the Residuals?

Suppose:

$$\mathbf{y} = (8, 10, 13).$$

Two candidate fits are:

$$\hat{\mathbf{y}}_A = (7, 11, 13), \quad \hat{\mathbf{y}}_B = (4, 14, 13).$$

Their residuals are:

$$\mathbf{e}_A = (1, -1, 0), \quad \mathbf{e}_B = (4, -4, 0).$$

But both residual sums are zero:

$$\sum_i e_{A,i} = 0, \quad \sum_i e_{B,i} = 0.$$

Squared Error Distinguishes the Fits

Use squared residuals instead:

$$\sum_i e_{A,i}^2 = 1^2 + (-1)^2 + 0^2 = 2.$$

For Model B:

$$\sum_i e_{B,i}^2 = 4^2 + (-4)^2 + 0^2 = 32.$$

Therefore:

$$2 < 32$$

so Model A is the better fit under squared error.

This motivates the residual sum of squares:

$$\text{RSS}(\beta) = \sum_{i=1}^n (y_i - \mathbf{x}_i^\top \beta)^2.$$

The Least-Squares Fitting Problem

For candidate coefficients β , define:

$$\text{RSS}(\beta) = \sum_{i=1}^n (y_i - \mathbf{x}_i^\top \beta)^2$$

Ordinary least squares chooses:

$$\hat{\beta} = \arg \min_{\beta} \text{RSS}(\beta)$$

Equivalently:

$$\hat{\beta} = \arg \min_{\beta} \|\mathbf{y} - \mathbf{X}\beta\|_2^2.$$

The next question is: how do we solve this optimization problem?

Least Squares and the OLS Solution

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How Do We Solve Least Squares?

For simple linear regression:

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i.$$

The least-squares objective is:

$$\text{RSS}(\beta_0, \beta_1) = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2.$$

There are two unknown coefficients:

$$\beta_0, \quad \beta_1.$$

At the minimum:

$$\frac{\partial \text{RSS}}{\partial \beta_0} = 0, \quad \frac{\partial \text{RSS}}{\partial \beta_1} = 0.$$

We solve these two equations in turn.

Step 1: Solve for the Intercept

Differentiate the loss with respect to β_0 :

$$\frac{\partial RSS}{\partial \beta_0} = -2 \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)$$

At the minimum:

$$\sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$$

Using:

$$\sum_i y_i = n\bar{y}, \quad \sum_i x_i = n\bar{x},$$

we obtain:

$$\boxed{\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}}$$

Step 2: Derive the Slope Equation

Differentiate:

$$\frac{\partial RSS}{\partial \beta_1} = -2 \sum_i x_i (y_i - \beta_0 - \beta_1 x_i)$$

At the optimum:

$$\sum_i x_i e_i = 0, \quad e_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i.$$

From Step 1:

$$\sum_i e_i = 0.$$

Therefore:

$$\sum_i x_i e_i - \bar{x} \sum_i e_i = 0$$

and:

$$\boxed{\sum_i (x_i - \bar{x}) e_i = 0}$$

Step 3: Solve for the Slope

From Step 1:

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

Hence:

$$e_i = (y_i - \bar{y}) - \hat{\beta}_1(x_i - \bar{x})$$

Substitute into:

$$\sum_i (x_i - \bar{x})e_i = 0$$

giving:

$$\sum_i (x_i - \bar{x})(y_i - \bar{y}) = \hat{\beta}_1 \sum_i (x_i - \bar{x})^2$$

Therefore:

$$\hat{\beta}_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}$$

Worked Example: Find the Regression Line

Consider:

$$x = (1, 2, 3), \quad y = (2, 3, 5).$$

The means are:

$$\bar{x} = 2, \quad \bar{y} = \frac{10}{3}.$$

Using:

$$\hat{\beta}_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2},$$

the numerator is:

$$(-1) \left(-\frac{4}{3}\right) + 0 \left(-\frac{1}{3}\right) + 1 \left(\frac{5}{3}\right) = 3,$$

and the denominator is:

$$(-1)^2 + 0^2 + 1^2 = 2.$$

Therefore:

$$\boxed{\hat{\beta}_1 = \frac{3}{2}}$$

Worked Example: Find the Intercept

Using:

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x},$$

we obtain:

$$\hat{\beta}_0 = \frac{10}{3} - \frac{3}{2}(2) = \frac{1}{3}.$$

Therefore the fitted regression line is:

$$\hat{y} = \frac{1}{3} + \frac{3}{2}x$$

For example, when:

$$x = 2,$$

the prediction is:

$$\hat{y} = \frac{1}{3} + 3 = \frac{10}{3}.$$

From Simple to Multiple Regression

The same idea extends to multiple features.

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

where:

$$\mathbf{X} \in \mathbb{R}^{n \times d}, \quad \boldsymbol{\beta} \in \mathbb{R}^d.$$

The least-squares objective is:

$$RSS(\boldsymbol{\beta}) = \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2$$

Equivalently:

$$RSS(\boldsymbol{\beta}) = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^\top (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$$

Now we derive the optimal coefficient vector.

Expanding the Objective

Expand:

$$RSS(\beta) = (\mathbf{y} - \mathbf{X}\beta)^\top (\mathbf{y} - \mathbf{X}\beta)$$

which gives:

$$= \mathbf{y}^\top \mathbf{y} - 2\beta^\top \mathbf{X}^\top \mathbf{y} + \beta^\top \mathbf{X}^\top \mathbf{X} \beta$$

Only the terms containing β affect the optimum.

The Normal Equations

Differentiate with respect to β :

$$\nabla_{\beta} RSS = -2\mathbf{X}^T \mathbf{y} + 2\mathbf{X}^T \mathbf{X} \beta$$

At the minimum:

$$\nabla_{\beta} RSS = 0$$

Therefore:

$$\boxed{\mathbf{X}^T \mathbf{X} \hat{\beta} = \mathbf{X}^T \mathbf{y}}$$

These equations characterize the OLS solution.

Closed-Form OLS Solution

If:

$$\mathbf{X}^T \mathbf{X}$$

is invertible, multiply both sides by its inverse:

$$\boxed{\hat{\beta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}}$$

This formula gives the unique OLS solution.

However, the inverse exists only when the features provide independent information.

When Is the Solution Unique?

A unique solution requires:

$$\text{rank}(\mathbf{X}) = d$$

If feature columns are linearly dependent:

$$\mathbf{X}\mathbf{a} = 0$$

for some nonzero vector $\mathbf{a}$.

Then:

$$\mathbf{X}(\boldsymbol{\beta} + \mathbf{a}) = \mathbf{X}\boldsymbol{\beta}$$

so different coefficients produce the same prediction.

The prediction can be unique even when the coefficients are not.

Example: Perfect Multicollinearity

Suppose $x_{i2} = 2x_{i1}$ for every observation. Then:

$$\beta_1 x_{i1} + \beta_2 x_{i2} = (\beta_1 + 2\beta_2)x_{i1}$$

Only the combination:

$$\beta_1 + 2\beta_2$$

can be identified.

For example:

$$(\beta_1, \beta_2) = (3, 0) \text{ and } (\beta_1, \beta_2) = (1, 1)$$

produce the same fitted contribution:

$$3x_{i1}.$$

Why Does Least Squares Have a Statistical Meaning?

Assume Gaussian noise:

$$Y_i = \mathbf{x}_i^\top \boldsymbol{\beta} + \varepsilon_i,$$

with:

$$\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$$

Then:

$$Y_i | \mathbf{x}_i \sim \mathcal{N}(\mathbf{x}_i^\top \boldsymbol{\beta}, \sigma^2)$$

The likelihood contains the term:

$$\exp\left(-\frac{(y_i - \mathbf{x}_i^\top \boldsymbol{\beta})^2}{2\sigma^2}\right)$$

Therefore maximizing likelihood is equivalent to minimizing squared errors.

Gaussian MLE and OLS Are Equivalent

For independent observations:

$$\ell(\beta, \sigma^2) = C - \frac{1}{2\sigma^2} \sum_i (y_i - \mathbf{x}_i^\top \beta)^2$$

where C does not depend on β .

Therefore:

$$\arg \max_{\beta} \ell = \arg \min_{\beta} \sum_i (y_i - \mathbf{x}_i^\top \beta)^2$$

Hence:

Gaussian MLE = OLS

Why Does Gaussian Noise Appear in OLS?

OLS itself does not require a Gaussian assumption:

$$\hat{\beta} = \arg \min_{\beta} \|\mathbf{y} - \mathbf{X}\beta\|_2^2$$

This is only an optimization criterion.

If we additionally assume:

$$\varepsilon_i \sim \mathcal{N}(0, \sigma^2),$$

then:

OLS = Maximum Likelihood Estimation

The Gaussian assumption provides:

- ▶ a probabilistic interpretation of squared error;
- ▶ coefficient uncertainty and classical inference.

How Should OLS Be Computed?

The formula:

$$(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

is useful for derivation.

However, numerical software usually avoids explicitly computing:

$$(\mathbf{X}^T \mathbf{X})^{-1}$$

because it may amplify numerical errors.

Common approaches:

QR decomposition	or	SVD
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Least Squares as Projection

The fitted values are:

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$$

Therefore:

$$\hat{\mathbf{y}} \in \text{Col}(\mathbf{X})$$

The regression model can only produce vectors inside this feature space.
OLS chooses the closest possible prediction:

$$\hat{\mathbf{y}} = \text{Proj}_{\text{Col}(\mathbf{X})}(\mathbf{y})$$

The remaining part is the residual:

$$\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}}$$

Why Is the Residual Orthogonal?

At the optimum:

$$\mathbf{X}^\top(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = 0$$

Because:

$$\mathbf{e} = \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}},$$

we have:

$$\boxed{\mathbf{X}^\top \mathbf{e} = 0}$$

Each column of $\mathbf{X}$ is a feature direction.

Therefore:

$$\boxed{\mathbf{e} \perp \text{Col}(\mathbf{X})}$$

The residual contains no remaining linear pattern that the model could explain.

The Hat Matrix: Mapping Data to Predictions

The fitted values are:

$$\hat{y} = X\hat{\beta}$$

Substituting the OLS solution:

$$\hat{y} = X(X^T X)^{-1} X^T y$$

Define:

$$H = X(X^T X)^{-1} X^T$$

Then:

$$\hat{y} = Hy$$

The hat matrix describes how the observed responses are transformed into fitted responses.

Why Is the Hat Matrix a Projection?

The hat matrix projects the response onto the space of possible predictions:

$$\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$$

OLS separates the response into:

$$\mathbf{y} = \hat{\mathbf{y}} + \mathbf{e}$$

where:

$$\hat{\mathbf{y}} \in \text{Col}(\mathbf{X})$$

is the explainable part, and $\mathbf{e}$ is the remaining residual.

Because $\hat{\mathbf{y}}$ is already in the prediction space:

$$\mathbf{H}\hat{\mathbf{y}} = \hat{\mathbf{y}}, \text{ or } \mathbf{H}^2 = \mathbf{H}$$

Three Views of Ordinary Least Squares

The OLS solution can be understood from three perspectives:

Perspective	Key idea
Optimization	find coefficients minimizing squared error
Probability	the MLE under Gaussian noise
Geometry	project $\mathbf{y}$ onto the feature space

All perspectives lead to the same estimator:

$$\hat{\beta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

Coefficient Interpretation and Statistical Uncertainty

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What Does a Regression Coefficient Mean?

For multiple regression:

$$E[Y|\mathbf{X} = \mathbf{x}] = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$$

Coefficient β_j describes the change in the conditional mean when x_j increases by one unit, while other features are fixed:

$$\beta_j = \frac{\partial E[Y|\mathbf{X} = \mathbf{x}]}{\partial x_j}$$

Example:

$$E[Y|x_1, x_2] = 20 + 3x_1 - 2x_2$$

Increasing x_1 by one unit changes the fitted mean by +3 while keeping x_2 unchanged.

Coefficient Values Depend on the Scale

Suppose:

Y = house price, x = area.

If:

$$\hat{\beta} = 150000,$$

then:

$1m^2 \rightarrow 150000$ HKD increase

If area is measured in units of $100m^2$ instead:

$$x' = x/100$$

the numerical coefficient changes.

Coefficient Estimates from Different Samples

The true model is:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

The OLS estimate is:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

Different training samples contain different noise:

$$\boldsymbol{\varepsilon}^{(1)}, \boldsymbol{\varepsilon}^{(2)}, \dots$$

Therefore the fitted coefficients also vary:

$$\hat{\boldsymbol{\beta}}^{(1)}, \hat{\boldsymbol{\beta}}^{(2)}, \dots$$

Coefficient uncertainty describes this variation across repeated samples.

Decomposition of the OLS Estimate

Substituting:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

into the OLS estimator gives:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T (\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon})$$

Therefore:

$$\hat{\boldsymbol{\beta}} = \boldsymbol{\beta} + (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \boldsymbol{\varepsilon}$$

The first term is the true coefficient.

The second term is the estimation error caused by noise.

Unbiasedness of OLS

Assume:

$$E[\epsilon|\mathbf{X}] = 0$$

Taking expectation:

$$E[\hat{\beta}|\mathbf{X}] = \beta + (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T E[\epsilon|\mathbf{X}]$$

Hence:

$$E[\hat{\beta}|\mathbf{X}] = \beta$$

OLS estimates are centered around the true coefficients under the zero conditional mean assumption.

From Coefficient Error to Uncertainty

The estimated coefficients vary because different training samples contain different noise.
From the OLS estimator:

$$\hat{\beta} - \beta = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \epsilon$$

The coefficient error is a linear transformation of the observation noise:

data noise $\longrightarrow$ coefficient uncertainty

Therefore, we study:

$$\text{Var}(\hat{\beta} \mid \mathbf{X})$$

Variance of the OLS Estimate

Assume independent errors with the same variance:

$$\text{Var}(\boldsymbol{\varepsilon} \mid \mathbf{X}) = \sigma^2 \mathbf{I}$$

Let:

$$\mathbf{A} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T$$

Since:

$$\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = \mathbf{A}\boldsymbol{\varepsilon}$$

the variance is:

$$\text{Var}(\hat{\boldsymbol{\beta}} \mid \mathbf{X}) = \mathbf{A}(\sigma^2 \mathbf{I})\mathbf{A}^T$$

Therefore:

$$\boxed{\text{Var}(\hat{\boldsymbol{\beta}} \mid \mathbf{X}) = \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1}}$$

Factors Affecting Coefficient Uncertainty

From:

$$\text{Var}(\hat{\beta}|\mathbf{X}) = \sigma^2(\mathbf{X}^T \mathbf{X})^{-1}$$

coefficient uncertainty depends on two sources:

Noise level σ^2	more unexplained variation increases uncertainty
Feature information	more data variation provides more information
Feature correlation	overlapping features make coefficients unstable

A model can predict well even when individual coefficients are uncertain.

Estimating the Noise Variance

The variance formula contains the unknown noise level:

$$\sigma^2$$

After fitting the model, we estimate it from residuals:

$$RSS = \sum_{i=1}^n e_i^2$$

The estimated noise variance is:

$$\hat{\sigma}^2 = \frac{RSS}{n - d}$$

where d is the number of fitted coefficients.

The term $n - d$ is the residual degrees of freedom.

Standard Error of a Coefficient

Replacing σ^2 with $\hat{\sigma}^2$:

$$\widehat{Var}(\hat{\beta}) = \hat{\sigma}^2 (\mathbf{X}^T \mathbf{X})^{-1}$$

For coefficient j :

$$SE(\hat{\beta}_j) = \hat{\sigma} \sqrt{[(\mathbf{X}^T \mathbf{X})^{-1}]_{jj}}$$

Therefore, a coefficient should be reported as:

$$\hat{\beta}_j \quad \text{with} \quad SE(\hat{\beta}_j)$$

From Normal to the t Distribution

Under Gaussian regression:

$$\hat{\beta}_j \sim \mathcal{N}(\beta_j, \sigma^2[(X^T X)^{-1}]_{jj})$$

If σ were known:

$$\frac{\hat{\beta}_j - \beta_j}{SE(\hat{\beta}_j)} \sim \mathcal{N}(0, 1)$$

In practice, σ is estimated from residuals.

Therefore:

$$\boxed{\frac{\hat{\beta}_j - \beta_j}{SE(\hat{\beta}_j)} \sim t_{n-d}}$$

Confidence Interval for a Coefficient

A $(1 - \alpha)$ confidence interval is:

$$\hat{\beta}_j \pm t_{1-\alpha/2, n-d} SE(\hat{\beta}_j)$$

Example:

$$\hat{\beta}_1 = 2.4, \quad SE = 0.6, \quad t_{0.975, 40} = 2.02$$

Therefore:

$$CI = 2.4 \pm 2.02(0.6)$$

$$CI_{0.95} = [1.19, 3.61]$$

Mean Response and Future Observation

For a new input x_0 , regression can answer two questions.

Mean response	What is the average outcome at x_0 ?
Future observation	What value will one new sample take?

The mean response is:

$$E[Y|x_0]$$

and uncertainty comes from estimating $\hat{\beta}$.

A future observation is:

$$Y_0 = x_0^T \beta + \varepsilon_0$$

which contains an additional random noise term:

$$\varepsilon_0$$

Uncertainty of the Mean Response

The estimated mean response is:

$$\hat{\mu}_0 = \mathbf{x}_0^T \hat{\boldsymbol{\beta}}$$

Its standard error is:

$$SE(\hat{\mu}_0) = \hat{\sigma} \sqrt{\mathbf{x}_0^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_0}$$

This describes uncertainty in:

$$E[Y|\mathbf{x}_0]$$

rather than a single future observation.

Prediction Uncertainty

A future observation contains:

$$Y_0 = \mathbf{x}_0^T \boldsymbol{\beta} + \varepsilon_0$$

Therefore prediction uncertainty includes:

uncertainty of fitted mean + new observation noise

The standard error becomes:

$$SE_{pred} = \hat{\sigma} \sqrt{1 + \mathbf{x}_0^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_0}$$

Confidence Interval and Prediction Interval

The mean-response interval uses:

$$\mathbf{x}_0^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_0$$

The prediction interval contains an additional noise term:

$$1 + \mathbf{x}_0^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_0$$

Therefore:

$$PI \text{ is wider than } CI$$

because predicting an individual outcome includes irreducible noise.

Prediction Uncertainty and Input Location

For a new input x_0 :

$$SE(\hat{\mu}_0) = \hat{\sigma} \sqrt{x_0^T (\mathbf{X}^T \mathbf{X})^{-1} x_0}$$

Consider a simple regression:

$$x \in [1, 5]$$

Prediction at $x_0 = 3$ may give:

$$SE(\hat{\mu}_0) = 0.5$$

while an extrapolated point $x_0 = 20$ may give:

$$SE(\hat{\mu}_0) = 3.0$$

The same model can be confident in one region and uncertain in another.

Summary of Regression Uncertainty

Regression provides two types of information:

$$\hat{\beta}_j$$

the estimated relationship,
and:

$$SE(\hat{\beta}_j)$$

the uncertainty of that estimate.
For prediction:

mean response $\neq$ future observation

because a future observation contains additional noise.

Diagnostics, Influence, and Generalization

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Why Model Diagnostics Matter

OLS finds:

$$\hat{\beta} = \arg \min_{\beta} \|\mathbf{y} - \mathbf{X}\beta\|_2^2$$

This gives the best linear fit to the observed data.

However, we still need to check:

- ▶ whether the relationship is approximately linear;
- ▶ whether the error behavior is reasonable;
- ▶ whether some observations dominate the fit;
- ▶ whether the model generalizes.

A fitted model is not automatically a trustworthy model.

Residuals as a Diagnostic Tool

After fitting:

$$e_i = y_i - \hat{y}_i$$

Residuals should not contain systematic patterns.

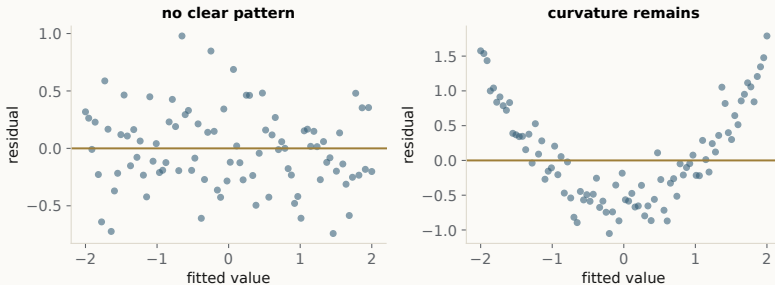
A useful diagnostic plot is:

$$e_i \text{ versus } \hat{y}_i$$

Common patterns:

Curvature	missing nonlinear structure
Fan shape	changing error variance
Extreme points	unusual observations

Checking the Linear Mean Structure



Random scatter suggests that the linear mean is reasonable.

A systematic curve suggests:

$$E[Y|X]$$

is not well captured by the current model.

Checking the Error Variance

Constant variance assumes:

$$\text{Var}(\varepsilon_i|x_i) = \sigma^2$$

A changing residual spread indicates:

$$\text{Var}(\varepsilon_i|x_i)$$

depends on the input.

This is called:

heteroskedasticity

It mainly affects uncertainty estimates.

Heteroskedasticity and Standard Errors

Under constant variance:

$$\text{Var}(\varepsilon|X) = \sigma^2 I$$

If variance changes:

$$\text{Var}(\varepsilon|X) \neq \sigma^2 I$$

The fitted coefficients remain:

$$\hat{\beta}_{OLS}$$

but the classical standard errors may become unreliable.
Robust standard errors adjust uncertainty estimation.

Unusual Inputs and Leverage

Some observations are unusual because of their feature values.

Example:

x_i is far from the region covered by other training examples. Such points have high leverage.

Leverage is measured by the diagonal of the hat matrix:

$$h_{ii} = H_{ii}$$

where:

$$\hat{y} = Hy$$

Leverage: Unusual Feature Positions

Some observations have unusual feature values.

These points can strongly affect the fitted model because their locations are far from the center of the training data.

The hat matrix measures this effect:

$$\hat{y} = Hy$$

with:

$$h_{ii} = H_{ii}$$

A large h_{ii} means that observation i has high leverage.

Example: Computing Leverage

For simple regression with an intercept:

$$h_{ii} = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{\sum_j (x_j - \bar{x})^2}$$

Consider: $x = (1, 2, 3, 4, 10)$. The mean is: $\bar{x} = 4$, and:

$$\sum_j (x_j - \bar{x})^2 = 60$$

For the last observation:

$$h_{55} = \frac{1}{5} + \frac{(10 - 4)^2}{60}$$

$$h_{55} = 0.80$$

Interpreting High Leverage

For the same dataset, $x = 1, 2, 3, 4$ are close to the center $\Rightarrow$ lower leverage
but $x = 10$ is far from the observed region $\Rightarrow h_{55} = 0.80$

This point has strong potential to change the fitted line.

Leverage depends on:

X

not on:

Y

so it measures unusual input location, not prediction error.

High leverage means the observation has strong influence potential because of its feature position.

Leverage and Residuals Describe Different Problems

Residuals and leverage measure different aspects:

Residual e_i Does the response disagree with the fitted model?

Leverage h_{ii} Is the input location unusual?

A point can have:

high leverage + small residual

or:

low leverage + large residual

Leverage measures feature geometry, not prediction error.

Influential Observations

Some observations can strongly change the fitted model.
Influence usually appears when:

high leverage + large residual

because:

- ▶ the point is unusual in feature space;
- ▶ the observed response disagrees with the model.

These observations deserve investigation.

Cook's Distance

Cook's distance combines residual size and leverage:

$$D_i \propto e_i^2 \frac{h_{ii}}{(1 - h_{ii})^2}$$

Large D_i means that removing the observation may noticeably change the fitted model.
However:

influential $\neq$ incorrect

An unusual observation may represent an important real case.

Multicollinearity and Coefficient Stability

Coefficient uncertainty depends on:

$$\text{Var}(\hat{\beta}) = \sigma^2(X^T X)^{-1}$$

If predictors contain overlapping information:

$$x_1 \approx x_2$$

the model has difficulty separating their individual effects.

As a result:

$$SE(\hat{\beta}_j)$$

can become large.

Example: Correlated Predictors

Suppose we predict house price using:

$$Price = \beta_0 + \beta_1(\text{area}) + \beta_2(\text{rooms})$$

Assume $\text{area} = 100m^2$ and $\text{rooms} = 3$. Both features describe house size, so they contain overlapping information. Two fitted models may give:

$$(\hat{\beta}_1, \hat{\beta}_2) = (2.0, 10), \text{ or } (\hat{\beta}_1, \hat{\beta}_2) = (2.5, -6)$$

with similar predictions.

The individual coefficients are unstable, but $\hat{y}$ can remain almost unchanged.

Multicollinearity makes feature effects difficult to separate.

Variance Inflation Factor

Multicollinearity occurs when predictors contain overlapping information. Suppose the model contains:

$$X_1, X_2, \dots, X_p$$

To evaluate predictor X_j , regress it on all remaining predictors:

$$X_j = \gamma_0 + \sum_{k \neq j} \gamma_k X_k + \epsilon$$

The coefficient of determination of this auxiliary regression is:

$$R_j^2$$

It measures how much of X_j can be explained by other predictors.

Interpreting R_j^2 in VIF

If:

$$R_j^2 \approx 0$$

other predictors cannot explain X_j .
The feature contains unique information.

If:

$$R_j^2 \approx 1$$

other predictors can almost reconstruct X_j .
The feature is highly redundant.

Variance Inflation Factor

The variance inflation factor is:

$$VIF_j = \frac{1}{1 - R_j^2}$$

Examples:

$$R_j^2 = 0 \Rightarrow VIF_j = 1$$

No multicollinearity.

$$R_j^2 = 0.9 \Rightarrow VIF_j = 10$$

Most information in X_j is shared with other predictors.

Overall Model Fit

After studying coefficient stability, we evaluate the response prediction.
The baseline model ignores all predictors:

$$\hat{y}_i = \bar{y}$$

Its total variation is:

$$TSS = \sum_i (y_i - \bar{y})^2$$

This is the error from always predicting the mean.

R^2 : Improvement over the Mean Baseline

The regression model has residual error:

$$RSS = \sum_i (y_i - \hat{y}_i)^2$$

Comparing with the baseline:

$$R^2 = 1 - \frac{RSS}{TSS}$$

Interpretation:

R^2 = fraction of response variation explained by the model

The Limitation of Training R^2

Adding predictors cannot increase training error:

$$RSS_{\text{new}} \leq RSS_{\text{old}}$$

Therefore:

$$R^2$$

always increases or stays unchanged on training data.

A useless feature may still improve training R^2 .

Adjusted R^2

Adjusted R^2 introduces a complexity penalty:

$$R_{\text{adj}}^2 = 1 - \frac{RSS/(n - d)}{TSS/(n - 1)}$$

where d is the number of fitted coefficients.

Unlike ordinary R^2 :

$$R_{\text{adj}}^2$$

can decrease when unnecessary predictors are added.

From Training Fit to Generalization

A model can fit the training data well but fail on unseen data.
Machine learning focuses on:

prediction performance on new observations

Therefore we separate:

D_{train}

for fitting and:

D_{test}

for evaluation.

Test Mean Squared Error

For test observations:

$$(x_i^{test}, y_i^{test})$$

evaluate:

$$MSE_{\text{test}} = \frac{1}{m} \sum_i (y_i^{test} - \hat{y}_i^{test})^2$$

Training error measures:

fit to observed data

Test error measures:

generalization ability

A Practical Regression Workflow

After fitting a linear regression model:

1. **Understand the relationship:** interpret coefficients and fitted values.
2. **Check residuals:** look for nonlinear patterns and changing variance.
3. **Check unusual observations:** use leverage and influence measures.
4. **Check feature redundancy:** examine coefficient stability and multicollinearity.
5. **Evaluate generalization:** measure performance on unseen data.

Regression analysis is not only fitting a line; it is checking whether the learned relationship can be trusted.

Summary

This lecture connected four questions in linear regression:

1. **Modeling:** How do we represent a numerical relationship?
2. **Optimization:** How do we estimate coefficients using least squares?
3. **Uncertainty:** How reliable are the estimated coefficients and predictions?
4. **Diagnostics:** Does the model describe the data and generalize to new observations?

A regression model is not only a fitted equation; it is a statistical model that must be interpreted and evaluated.

Week 4: connect the statistical question, mathematics, and evidence

Questions and discussion